

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 1 of 8

(1) Research Program/Project/Study Title: ACCEPTABILITY OF NIPA SAP SYRUP AND FRUIT JAM	
Proponent Name:	JANET V. LEDESMA
Academic Rank:	Asst. Professor III
Co-Authors (with academic rank):	LOUELLA P. DADIVAS
Campus/Satellite College/Address:	CAPSU DAYAO SATELLITE COLLEGE
Telephone/Cell Phone/Fax Number:	0921 511 7261
E-mail Address:	jvledesma@capsu.edu.ph

Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
---	--

(4) Site of implementation	
Municipality/City:	ROXAS CITY
Province:	CAPIZ
Region:	VI

(5) Classification Research: ☒ Basic ☐ Applied Development: ☐ Pilot Testing ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
---	---

(7) Sector/Commodity Food Processing Sector	(8) Discipline/Field of Specialization Food and Beverages Services
---	--

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 2 of 8

(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, the thrusts being addressed, utilization of the expected output, and the impact information generated will have on science, the target users, and the country.)

Nypa fruticans, also known as nipa palm or nipa, belongs to the Arecaceae family. They are a palm native to coastlines of the Indian and Pacific Oceans, it is considered to be the only palm adapted to the mangrove biome (Nipa Palm, n.d.). Nipa is a monoecious plant with a stout, trunkless, thornless rootstock with parts of it submerged in water or mud.

Nipa are often found near rivers and mangroves. It has large, rosette-arranged leaves that can reach 5 to 10 meters in length. Fruits from the nipa palm grow in clusters when young, its seeds are soft, juicy and edible which are like coconut meat. According to Ulyarti, et al, Fresh nipa fruits have a high percentage of water that decreases when it matures and a high moisture content. It is also considered as a delicacy in southeast asia. Its peduncle can be tapped for Nipa sap, in which the sap could be used for vinegar, wine, syrup, fermented juices and as sugar (Mateo et al, 2023)

Plantation area, processing, species, and harvesting time are the main determinants of the nutritional quality and sugar profile of palm syrup. Nipa palms are normally grown in wetland environments with mild tidal influences, therefore the nutrients in the soil and water have a significant impact on the sap composition and growth characteristics of the plant. (Saengkrajang, Chaijan, & Panpipat, 2021)

The utilization of Nipa fruit and sap faces various challenges and underexplored areas, hindering their optimal utilization and potential benefits. One significant issue is the limited commercialization, which is attributed to a lack of awareness, insufficient marketing efforts, and a limited understanding of their potential applications across industries. Additionally, the culinary applications of Nipa fruit and sap remain underexplored, with a gap in research and development for innovative uses in popular food products. Cultural and traditional practices in some regions have not evolved to meet contemporary demands, limiting their broader acceptance. Insufficient awareness of the nutritional and health benefits further creates a gap in consumer interest, necessitating research to highlight these aspects. Unsustainable harvesting practices pose a threat to Nipa palm ecosystems, impacting the availability of fruit and sap, highlighting a need for promoting sustainable harvesting methods. The lack of standardization in processing and quality control introduces variations in products, potentially hindering wider market acceptance. Addressing these challenges requires interdisciplinary research, collaboration with local communities, sustainable practices development, and promotional efforts to increase awareness and appreciation for the diverse uses of Nipa fruit and sap.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

The main purpose of the study was to evaluate the economic value, acceptability and best cooking temperature over time of nipa sap and fruit jam. The purpose of this research will determine the most acceptable cooking temperature of Nipa Sap Syrup-Fruit Jam production from nipa sap and Nipa Fruits by optimized heating process. The treatment of heating will start at 100 °C until the sap concentrated, followed by increasing temperature to 120 °C and 140 °C until the Nipa Sap Syrup-Fruit Jam reached the preferred method to Jam production.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

1. To determine the most acceptable treatment of Nipa Sap Syrup-Fruit Jam
2. To determine the cooking temperature of Nipa Sap Syrup-Fruit Jam of the most acceptable treatment.
3. To determine the sensory characteristic of Nipa Sap Syrup-Fruit Jam of the most acceptable treatment
4. To determine the consumers acceptability of the most acceptable treatment
5. Determine the Physicochemical Characteristics of the most acceptable characteristics (TSS, pH, and Viscosity) and microbial analysis (Yeast and Mold Count) of the most acceptable treatment
6. Determine the Nutritional Facts of Nipa Sap Syrup-Fruit Jam of the most acceptable treatment
7. Cost Analysis of Nipa Sap Syrup-Fruit Jam of the most acceptable treatment

	Document Type:	Document Code	RES-F09
		Revision No.	00
	Document Title:	Effective Date	January 26, 2023
		Page	Page 3 of 8

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

PH22016000606U1 NIPA (NYPa FRUITICANS) FRUIT JAM

This utility model disclosed a nipa (Nypa fruiticans) fruit jam as dessert or filling for baked goods. A fruit jam comprising a fresh, young nipa fruit and refined sugar in a ratio of 1:1 and 0.1 pcnt of citric acid based on the weight of the pulped used. The utility model also disclosed the process of making a nipa fruit jam.

PH22018000230U1 METHOD OF PRODUCING NIPA (Nypa fruticans) FRUIT MARMALADE

Disclosed is a method of preparing Nipa (Nypa fruticans) fruit marmalade, comprising the steps of washing the Nipa fruit to remove dirt and mud, sanitizing the fruit to remove dirt and mud, sanitizing the fruit by soaking to 100 ppm chlorinated water for 10 minutes, removing Nipa fruit from the water, removing the Nipa fruit from water, cutting the Nipa fruit into halves, scooping out the pulp, chopping the Nipa pulp, mixing the pulp with equal amount of sugar while stirring Constantly, heating the mixture up to 100 oc until it can be spooned out, pouring in a sterilized jar, sealing, labeling and coding.

(12) Scientific Basis/Theoretical Framework (Include concepts. theories and variables that indicate relevant relationships to be tested.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Materials

- Candy Thermometer
- Pan
- Preparation Bowl
- Wooden Spoon
- Weighing Scale
- Food Processor
- Knife
- Cheese Cloth

Experimental Treatment

A. Acceptability of Nipa Sap Syrup-Fruit Jam

	Treatment A	Treatment B	Treatment C	Treatment D
Most Acceptable Treatment	100	75%	50%	25%
		25%	50%	75%

B. Cooking Temperature

Ingredients	Treatment A	Treatment B	Treatment C	Treatment D
Nipa Syrup	100 °C	120 °C	140 °C	160 °C

Methodology

Making Nipa Syrup

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 4 of 8

1. Collecting Nipa Sap
2. Strain the Nipa Sap in cheese cloth.
3. Simmer at low temperature.
4. Stir continuously the nipa sap in an open-pan at 100 degrees Celsius until it reaches the syrup consistency.
5. Cool down at room temperature (27–30 °C) and keep at 4–10 °C until used.

Preparation of Nipa Fruit

1. Collect Nipa fruits. These are small, round fruits that grow in clusters.
2. Clean and Prepare: Wash the Nipa fruits thoroughly and remove any debris or unwanted parts.
3. Cut the nipa fruit and remove the meat inside the shell.
4. Collect and wash the nipa fruit.
5. Blender the meat using a food processor.

Preparation of Jam

1. Cook the Nipa fruit meat over low to medium heat. Stir continuously to prevent sticking.
2. Add Nipa Syrup.
3. Add flavor enhancers such as kalamansi juice and salt to brighten the taste.
4. Cook and stir continuously until reach 120 C to Jam Consistency
5. Allow the mixture to simmer until it thickens and reaches a jam-like consistency.
6. Cool and Store Let the Nipa Sap Syrup- fruit jam cool down.
7. Once cooled, transfer it into clean, sterilized jars for storage.

Color

Scoring	Verbal Interpretation/Response Categories	Scoring Interval
5	Golden Brown	4.2-5.0
4	Brown	3.4-4.9
3	Slightly brown	2.6-3.39
2	Moderately brown	1.8-2.59
1	yellow	1.0-1.79

For Odor

Scoring	Verbal interpretation/Response Categories	Scoring Interval
5	Extremely Sweet and Fruity	4.2-5.0
4	Moderately Sweet and Fruity	3.4-4.9
3	Neither Sweet and Fruity	2.6-3.39
2	Slightly Sweet and Fruity	1.8-2.59
1	Sweet and Fruity	1.0-1.79

For Texture

Scoring	Verbal Interpretation/Response Categories	Scoring Interval
5	Extremely Coarse and Thick	4.2-5.0
4	Moderately Coarse and Thick	3.4-4.9
3	Neither Coarse and Thick	2.6-3.39
2	Slightly Coarse and Thick	1.8-2.59
1	Coarse and Thick	1.0-1.79

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 6 of 8

3. Sensory Assessment/Organoleptic test													
4. Proximate analysis													
5. Microbial analysis													
6. Preparation & writing of data and results													
7. Presentation of Results and Recommendations													
8. IP Protection Application													

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

A nutritious condiment that is sweet and flavorful spread typically made from nipa syrup and fruit. It Introduces a unique and novel flavor profile using Nipa syrup as sweetener. Promoting the nutritional benefits of Nipa Jam, including its high water content, essential nutrients and potential health benefits.

(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Process will be taught to the target clientele through extension program of the college, however, prior to the extension activity, formulation of the Nipa syrup and fruit will be copyrighted.

(17) Target Beneficiaries (Who the clienteles are and what are the expected outcome/effects of the use of the projects output)

Small-Scale Farmers: Those engaged in Nipa palm cultivation and sap extraction can benefit from insights into the potential value addition of their produce, expanding their market reach.

Processing Companies: Companies involved in the production of Nipa sap syrup and fruit jam can gain valuable information for optimizing processing methods, enhancing product quality, and meeting market demand.

Community Members: Residents in areas where Nipa palms are prevalent may benefit from increased economic opportunities through the production and sale of Nipa sap products. The study can help empower local communities by promoting sustainable practices and community engagement.

Health-Conscious Consumers: Individuals seeking healthier and natural alternatives to conventional sweeteners and jams can benefit from a greater understanding of the nutritional properties and potential health benefits of Nipa sap syrup and fruit jam.

Agricultural Departments: Government bodies responsible for agriculture can use the study to promote sustainable practices in Nipa palm cultivation, ensuring environmental conservation and responsible harvesting.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 7 of 8

Researchers in the fields of agriculture, food science, and nutrition can use the study's findings to contribute to academic knowledge and further research on Nipa sap products.

(18) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(19) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount(PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	1	3,000
Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
Transportation expense		500.00
Physicochemical analysis	2	2,785.00
Packaging		1,500.00
TOTAL		4,,785.00
CAPITAL OUTLAY		
Nipa Sap	10cont.@ 600	6,000.00
Nipa Fruit	40 pcs. @ 60	2,400.00
Rock Salt	2kl @ 50	100.00
Kalamansi	2 kg@ 80	160.00
TOTAL		8,660.00
OVERALL TOTAL		13,445.00

(20) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

PH22018000610U1 A COMPOSITION FOR MUD CLAMS (ANODONTIA EDENTULA) SALAD (SISIG) WITH COCONUT MILK

PH22019001306U1 PROCESS OF PRODUCING MUSHROOM SISIG

PH22016000709U1 FORMULATION OF MILKFISH (CHANOS CHANOS) DUMPLING FILLING

(21) Capsule Curriculum Vitae (One-page brief C.V. only)



	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 8 of 8

JANET V. LEDESMA

Villa Alegria Subdivision
Punta Tabuc, Roxas City
Capiz, Philippines, 5800

Email: ledesmajanet35@yahoo.com | jvledesma@capsu.edu.ph

Contact #: 09215117261

EDUCATIONAL BACKGROUND

Doctoral Degree : Central Philippine University 2020-2021
Jaro, Iloilo
Doctor of Tourism and Hospitality Management

Master Degree : Colegio de la Purisima Concepcion 2010
Roxas City, Capiz
Master of Business Administration (M.B.A.)

College : Colegio Del Sagrado Corazon de Jesus 1996
Iloilo City
Bachelor of Science in Hotel and Restaurant Management

Work Experience:

Hercor College: Faculty 2005-2015

Subject Handle: Asian Cuisine, International Cuisine, Food and Beverage, Catering Management, Costing Management, Bakery Management, Total Quality Management, Bread and Pastry.

General Manager: Hercor Catering Services 2005-2015

Capiz State University: Professor 2015 to present
Subject Handle: Bakery Management, Food Safety, Principles to Human Nutrition, Entrepreneurship, Trade and Marketing

Gabriel's Catering and Services-owner 2015 to present
Gabriel's Event Center-owner
Rafael's Cakes-owner
Village Coffee-Owner